

Proposal #25

Remote Eye-Tracking *Software Glossary*

CS Capstone Team 12



PFW CS Department -Sponsor

Prof. Jay Johns -Faculty Advisor

Dominic Minnich -Team Leader

Sulaiman Hussain, Logan Smith, Kyle Benich -Team Members

Glossary

Admin

A user role with the highest permissions, responsible for managing user accounts, projects, and system settings.

Analyst

A user role focused on analyzing project data and providing insights, typically without permissions to create or manage projects.

AES-256 Encryption

A secure encryption standard used for protecting data at rest and in transit, ensuring the confidentiality of user information.

Breadcrumb Navigation

A user interface feature that displays the user's current location within a web application and provides links to previous sections for easy navigation.

Computer Vision

Technology used for real-time tracking of eye movements through standard webcams, analyzing gaze points and behavior without specialized hardware.

Data Encryption

The process of encoding data to prevent unauthorized access, crucial for ensuring user privacy and compliance with security protocols.

Eye-Tracking

A method for tracking and analyzing eye movements to study user interactions with digital interfaces; in this project, it is implemented via standard cameras.

Figma Domain

A reference to the Figma design platform; the project initially supports eye-tracking analysis specifically on this domain.

Flask

A lightweight web application framework used to develop the server backend for this eye-tracking tool.

Guest User

A limited-access user role that allows viewing shared projects without the ability to edit or create content.

Heat Mapping

A visual representation of areas where users' attention is focused on a screen, often used to analyze popular gaze points.

JavaScript, HTML, CSS

Core technologies used to build the front-end of the application, providing structure, style, and interactivity.

Mobile Support

Extending application capabilities to mobile devices, using embedded cameras for eye-tracking.

Multi-Factor Authentication (MFA)

A security feature requiring multiple forms of verification before granting access, enhancing login security.

Privacy Consent

A protocol where users agree to terms of service and grant necessary permissions, ensuring legal compliance for data collection.

Project Manager

A user role responsible for creating, sharing, and analyzing UX testing projects within the system.

Real-Time Eye-Tracking Overlay

A feature that visually represents users' gaze points on the screen during live sessions, aiding UX analysis.

Repository (GitHub)

A version-controlled codebase storage, which includes the project's history, source code, and documentation.

Role-Based Access Control (RBAC)

A system for defining permissions based on user roles, such as Admin, Project Manager, Analyst, and Guest, to maintain structured access.

Screen Recording

The process of capturing video of the user's screen activity along with eye-tracking data for analysis.

SQLite

A lightweight database used for managing and storing user data, project data, and analysis results.

Standard Webcam or Phone Camera

The hardware requirement for the application's eye-tracking feature, providing accessibility without specialized devices.

Success and Failure Conditions

Criteria defining successful or unsuccessful outcomes for specific system interactions, like password resets.

Testing and Validation

Processes including unit tests, user acceptance testing, and scenario-based testing to ensure software reliability and performance.

User Acceptance Testing (UAT)

A testing phase where end-users validate the software's functionality and usability before full deployment.

User Behavior Insights

Data-driven findings about user interaction patterns, aiding in UX optimization and design improvements.

User Experience (UX)

The overall experience and satisfaction a user has when interacting with a digital product or system, which this tool aims to enhance through eye-tracking.

UX Testing

The process of evaluating how users interact with a product to identify usability issues, improve functionality, and enhance the overall user experience.

Webgazer.js

A JavaScript library used for implementing eye-tracking features through browsers, capturing gaze data without specialized hardware.

WCAG 2.1 Level AA Accessibility Guidelines

Web Content Accessibility Guidelines ensuring the application is accessible to all users, including those with disabilities.

Wireframes

Preliminary visual representations of the application's user interface, guiding design decisions for usability.